Project ECHO: Comprehensive Mathematical & Tabular Reference

Documentation Suite

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1 Introduction

This document serves as the mathematical and tabular reference for **Project ECHO** (Emotionally Cognizant Heuristic Orchestrator). It outlines the core algorithms used for feature extraction, the test suite thresholds for the state machine, the Dynamic Difficulty Adjustment (DDA) parameters for the games, and the evaluation metrics equations used to grade the system.

2 Mathematical Concepts & Feature Extraction

2.1 Cognitive Load Complexity (CLC)

Cognitive load is estimated from keystroke dynamics using the Inter-Key Interval (IKI). The CLC metric is normalized based on the mean and variance of the IKI distribution over a 2-second rolling window.

$$\text{CLC} = \alpha \left(\frac{\mu_{\text{IKI}}}{\sigma_{\text{IKI}} + \epsilon} \right) + \beta \cdot \text{KPS} \tag{1}$$

Where μ_{IKI} is the mean inter-key interval, σ_{IKI} is the variance, KPS is Keys Per Second, and α, β are scaling weights.

2.2 Sigma-Lognormal Model (SLM) for Neuromotor Jitter

Mouse trajectories are modeled as a sequence of lognormal velocity profiles. Stress and frustration increase neuromotor jitter, reducing the Signal-to-Noise Ratio (SNR) of the trajectory.

$$v(t) = \sum_{i=1}^{N} \frac{D_i}{\sqrt{2\pi}\sigma_i(t - t_{0i})} \exp\left(-\frac{(\ln(t - t_{0i}) - \mu_i)^2}{2\sigma_i^2} \right) \tag{2}$$

Where D_i is stroke amplitude, t_{0i} is stroke onset time, μ_i is log-time delay, and σ_i is log-response time.

2.3 POMDP Belief Update

The emotional state is tracked as a Partially Observable Markov Decision Process. The belief state $b(s)$ is updated given an observation o (e.g., facial cues, DSTER classification).

$$b'(s') = \eta \cdot O(o \mid s') \sum_{s \in S} T(s' \mid s, a) b(s) \tag{3}$$

Where η is a normalizing constant, $O(o|s')$ is the observation probability, and $T(s'|s, a)$ is the transition probability.

3 Test Suite & State Machine Thresholds

The state machine drives the transitions based on strict temporal thresholds to prevent flickering and ensure reliable DDA updates.

State	Trigger Condition	Duration Threshold	Outcome / Fallback
`focused`	Game-bound Keys (WASD, Space, Mouse)	Instant	Increases game difficulty gradually
`joy`	Continuous smiling (MediaPipe Face)	10 seconds	Rewards player, eases tension
`angry`	Non-game Keys (Keyboard Bashing)	Instant	Drops obstacles or penalty prompts
`thinking`	Hand on chin (MediaPipe Hands)	3 seconds	Spawns contextual hints after 15s
`given_up`	Absolute zero input (AFK)	30 seconds	Red warning banner, pauses pacing
`neutral`	Sparse or ambiguous input	15 seconds	Baseline state, stabilizes DDA

Table 1: Project ECHO Live Demo State Machine Transitions and Thresholds.

4 Dynamic Difficulty Adjustment (DDA) Parameters

The parameter modulator affects four different game engines synchronously every 15 seconds based on the active POMDP state.

5 Evaluation Metrics Equations

When the session ends, the pipeline cross-references the true state machine indices (y_{true}) against the synthetic high-confidence evaluation array (y_{pred} and y_{scores}).

5.1 Standard Classification Metrics

$$\text{Precision} = \frac{TP}{TP + FP} \tag{4}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{5}$$

$$\text{F1 Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{6}$$

5.2 Cohen's Kappa

Cohen's Kappa (κ) measures inter-rater agreement for categorical items, correcting for chance agreement.

$$\kappa = \frac{p_o - p_e}{1 - p_e} \tag{7}$$

Where p_o is the relative observed agreement (Accuracy), and p_e is the hypothetical probability of chance agreement based on marginal probabilities.

Game Genre	Modulated Parameters (Range)
Left 4 Dead	• `enemy_spawn_rate` $(0.1 - 1.0)$ • `horde_frequency` $(0.1 - 1.0)$ • `health_item_scarcity` $(0.2 - 0.9)$ • `special_infected_cap` $(1.0 - 6.0)$ • `wanderer_density` $(0.1 - 1.0)$
Alien: Isolation	• `alien_menace_gauge` $(0.0 - 1.0)$ • `tension_budget` $(0.2 - 2.0)$ • `alien_search_radius` $(0.1 - 1.0)$ • `vent_drop_probability` $(0.1 - 0.8)$ • `noise_sensitivity` $(0.1 - 1.0)$
Resident Evil 4	• `action_score` $(1.0 - 10.0)$ • `enemy_aggression` $(0.2 - 1.0)$ • `enemy_accuracy` $(0.1 - 1.0)$ • `damage_taken_mult` $(0.5 - 2.0)$ • `item_drop_quality` $(0.1 - 1.0)$
Silent Hill	• `world_decay_speed` $(0.1 - 1.0)$ • `puzzle_obfuscation` $(0.1 - 1.0)$ • `static_interference` $(0.0 - 1.0)$ • `monster_type_bias` $(0.0 - 1.0)$ • `environment_lighting` $(0.1 - 1.0)$

Table 2: DDA parameter matrix managed by the Heuristic Orchestrator.